Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **C++/Python system software**, **Linux-based automation**, and robust **performance/debugging workflows** for high-reliability, sensor-driven, and distributed engineering systems.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving backend observability and reliability in **Linux**-based runtime environments.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving issue triage speed and system stability across distributed services and cross-functional engineering operations.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, integrating secure APIs with concurrent processing while improving debug visibility, telemetry traces, and repeatable validation coverage in production.
- Improved release velocity with **CI/CD** controls and automated checks, reducing turnaround time by **60%** while preserving quality through maintainable tests, root-cause analysis, and structured issue-resolution workflows.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling reproducible behavior analysis and performance diagnostics across safety-sensitive scenarios requiring disciplined benchmark execution and technical instrumentation.
- Produced technical summaries and debugging diagnostics that converted research findings into implementation plans, improving **defect isolation speed** and collaboration across software, systems, and model-evaluation teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations, dependable data exchange, and system-level performance tuning.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving service reliability and cross-service communication under multi-process, data-intensive workloads.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through script-driven validation and robust rollback mechanisms.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash
Systems & Embedded: Linux/UNIX, Multi-threading, Process/Memory Management, Real-time Concepts
Networking & Interfaces: TCP/IP, HTTP, SSH, WebSockets, Sensor/Data Integration Workflows
Tooling: Git, Jenkins, Docker, Kubernetes, CMake/Make, GDB-style Debugging, CI/CD
Engineering Practices: Unit/Integration Testing, Troubleshooting, Performance Tuning, Agile/Scrum

Projects

CPU Scheduling Simulator | *C++, React.js, JavaScript*

- Implemented FCFS, SJF, SRT, RR, and MLQ scheduling algorithms with visualization and metrics tooling, reducing debugging time by **20%** while strengthening low-level **C++** reasoning and performance analysis practices.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*

- Built a scalable data platform supporting **50K+ users** and **1M+ transactions** with secure integrations, real-time synchronization, and **Kubernetes**-backed deployments focused on reliability and throughput stability.